Karna Gowda

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PROFESSIONAL EXPERIENCE

The Ohio State University, Columbus, OH — 2024 – present
Assistant Professor, Department of Microbiology and Biophysics Graduate Program

The University of Chicago, Chicago, IL — 2020 – 2023
Postdoctoral Scholar, Department of Ecology & Evolution
Mentors: Seppe Kuehn and Madhav Mani

University of Illinois Urbana-Champaign, Urbana, IL — 2017 – 2020
James S. McDonnell Foundation Postdoctoral Fellow, Department of Physics
Mentors: Seppe Kuehn and Madhav Mani

EDUCATION

Northwestern University, Evanston, IL — 2012 – 2017
PhD, Engineering Sciences and Applied Mathematics
Advisor: Mary Silber

Marine Biological Laboratory, Woods Hole, MA — 2016
MBL Physiology Course

Northwestern University, Evanston, IL — 2011 – 2012
MS, Engineering Sciences and Applied Mathematics

University of Illinois Urbana-Champaign, Urbana, IL — 2005 – 2008
BS, Mathematics (with distinction)

PUBLICATIONS

*co-first author, †co-corresponding author.

12. K. Crocker, K. Lee, M. Chakraverti-Wuerthwein, Z. Li, M. Tikhonov, M. Mani, **K. Gowda**†, and S. Kuehn†. "Environmentally dependent interactions shape patterns in gene content across natural microbiomes." *Nat Microbiol* **9**, 2022–2037 (2024). doi:10.1038/s41564-024-01752-4

11. A. Skwara, **K. Gowda**, M. Yousef, J. Diaz-Colunga, A. Raman, A. Sanchez, M. Tikhonov, and S. Kuehn. "Statistically learning the functional landscape of microbial communities." *Nat Ecol Evol* **7**, 1823–1833 (2023). doi:10.1038/s41559-023-02197-4

    ↪ Highlighted by D. Amor. *Nat Ecol Evol* **7**, 1754–1755 (2023). doi:10.1038/s41559-023-02214-6

10. J. Diaz-Colunga*, A. Skwara*, **K. Gowda***, R. Diaz-Uriarte*, M. Tikhonov*, D. Bajic*, and A. Sanchez*. "Global epistasis on fitness landscapes." *Philos T R Soc B* **378**, 20220053 (2023). doi:10.1098/rstb.2022.0053

9. **K. Gowda** and S. Kuehn. "Microbial biofilms: An ecological tale of Jekyll and Hyde." *Curr Biol* **32**, R1349–R1351 (2022). doi:10.1016/j.cub.2022.10.068

8. **K. Gowda**, D. Ping, M. Mani, and S. Kuehn. "Genomic structure predicts metabolite dynamics in microbial communities." *Cell* **185**, 530–546 (2022). doi:10.1016/j.cell.2021.12.036

    ↪ Highlighted by A. Flamholz and D. Newman. *Curr Biol* **32**, R215–R218 (2022). doi:10.1016/j.cub.2022.02.002

7. C. Gopalakrishnappa*, **K. Gowda***, K. Prabhakara*, and S. Kuehn. "An ensemble approach to the structure-function problem in microbial communities." *iScience* **25**, 103761 (2022). doi:10.1016/j.isci.2022.103761

6. D. T. Fraebel, **K. Gowda**, M. Mani, and S. Kuehn. "Evolution of generalists by phenotypic plasticity." *iScience* **23**, 101678 (2020). doi:10.1016/j.isci.2020.101678

5. P. Gandhi, L. Werner, S. Iams, **K. Gowda**, and M. Silber. "A topographic mechanism for arcing of dryland vegetation bands." *J R Soc Interface* **15**, 20180508 (2018). doi:10.1098/rsif.2018.0508
4. **K. Gowda**, S. Iams, and M. Silber. "Signatures of human impact on self-organized vegetation in the Horn of Africa." *Sci Rep* **8**, 3622 (2018). doi:10.1038/s41598-018-22075-5
3. **K. Gowda**, Y. Chen, S. Iams, and M. Silber. "Assessing the robustness of spatial pattern sequences in a model of dryland vegetation." *Proc R Soc A* **472**, 20150893 (2016). doi:10.1098/rspa.2015.0893
2. **K. Gowda*** and C. Kuehn*. "Early-warning signs for pattern-formation in stochastic partial differential equations." *Commun Nonlinear Sci Numer Simul* **22**, 55–69 (2015). doi:10.1016/j.cnsns.2014.09.019
1. **K. Gowda**, H. Riecke, and M. Silber. "Transitions between patterned states in vegetation models for semiarid ecosystems." *Phys Rev E* **89**, 022701 (2014). doi:10.1103/PhysRevE.89.022701

GRANTS, HONORS & AWARDS

Postdoctoral

James S. McDonnell Foundation Postdoctoral Fellowship (# 220020499) — 2017 – 2020
"Evolving communities: how adaptation shapes microbial interactions."
Amount: $200,000

Graduate

NSF-RTG Graduate Training Fellowship, Northwestern University	2016 – 2017
NIH & Helmsley Charitable Trust Scholarship, Marine Biological Laboratory	2016
Presidential Fellowship Finalist, Northwestern University	2015
SAMSI Visiting Graduate Fellow, Program on Mathematical and Statistical Ecology	2014 – 2015
Professional Development Grant, The Graduate School, Northwestern University	2014
ComSciCon-Chicago Workshop Funding, Graduate Student Council, University of Chicago	2014
Walter P. Murphy Fellowship, Northwestern University	2012 – 2013
Travel grants to NSF Math & Climate Research Network meetings at SAMSI and Bowdoin College, and ComSciCon at Harvard University	2012 – 2014

Undergraduate

National Merit Scholarship	2005 – 2008
University of Illinois Honors Scholarship	2005 – 2006

SELECTED PRESENTATIONS

Invited talks and seminars

Biophysics Graduate Program Seminar, The Ohio State University, Columbus, OH.	September 11, 2024
Tara Oceans Retreat, Nice, France.	May 28, 2024
NITMB Ecological Dynamics of Microbial Communities Workshop, Chicago, IL.	April 29, 2024
Annual Microbial Communities Symposium, Columbus, OH.	April 12, 2024
American Physical Society March Meeting, Minneapolis, MN.	March 4, 2024
Annual Microbiology Symposium, Columbus, OH.	December 8, 2023
Microbial Sciences Institute Seminar, Yale University, West Haven, CT.	March 9, 2023
Ecology & Evolutionary Biology Seminar, Yale University, New Haven, CT.	March 8, 2023
Microbiology & Immunology Seminar, University of British Columbia, Vancouver, Canada.	March 1, 2023
Microbiology Seminar, Ohio State University, Columbus, OH.	February 20, 2023

- Microbiology & Immunology Seminar, Stanford University School of Medicine, Stanford, CA. — February 6, 2023
- College of Engineering Seminar, Boston University, Boston, MA. — January 26, 2023
- School of Life Sciences Seminar, Arizona State University, Tempe, AZ. — January 17, 2023
- Agricultural Biology Seminar, Colorado State University, Ft. Collins, CO. — December 5, 2022
- Advanced Biomedical Computation Seminar, Harvard Medical School (virtual). — October 24, 2022
- 18th International Symposium on Microbial Ecology (ISME18), Lausanne, Switzerland — August 16, 2022
- Laboratory of Living Matter, Institute for Advanced Study (virtual). — March 9, 2022
- Host-microbe Center Seminar, University of Oregon (virtual). — January 31, 2022
- KITP Program on the Ecology and Evolution of Microbial Communities, Santa Barbara, CA. — July 28, 2021
- Math, Statistics, and Computer Science Seminar, St. Olaf College, Northfield, MN. — September 13, 2019
- Institute for Genomic Biology Seminar, University of Illinois at Urbana-Champaign, Urbana, IL. — March 29, 2019
- Statistics Seminar, University of Chicago, Chicago, IL. — February 3, 2017
- Physics Seminar, University of Illinois at Urbana-Champaign, Urbana, IL. — April 8, 2016

Contributed talks and posters

- American Physical Society March Meeting, Chicago, IL. — March 16, 2022
- American Physical Society March Meeting (virtual). — March 3, 2020
- SIAM Conference on Dynamical Systems, Snowbird, UT. — May 23, 2017
- SIAM Conference on Dynamical Systems, Snowbird, UT (poster). — May 19, 2015
- Spatio-Temporal Dynamics in Ecology, Leiden, Netherlands (poster). — December 8, 2014
- International Centre for Mathematical Sciences Tipping Points, Edinburgh, U.K. (poster). — September 9, 2013
- SACNAS National Conference, San Antonio, TX. — October 5, 2013
- SIAM Conference on Dynamical Systems, Snowbird, UT. — May 19, 2013
- IUGG Conference on Mathematical Geophysics, Edinburgh, U.K. (poster). — June 18, 2012

TEACHING EXPERIENCE

Instructor

- MICRBIO 5130, Biology by the Numbers — 2025 – present
- MICRBIO 7899, Microbiology Colloquium at the Ohio State University — 2024 – present

Pedagogical service

- Guest lecturer for Quantitative Microbial Ecology course at University of Chicago — 2022
- Guest lecturer for Mathematical Biology course at St. Olaf College — 2020
- Developed a research module for Macalester College/Harvey Mudd College REU — 2016
- Developed and guest lecturer for *Ecological Resilience* module in Biomathematics course at Bowdoin College — 2013
- Peer instructor for Calculus and Linear Algebra preliminary prep sessions — 2012 – 2013

Teaching assistant

- *Physical Biology of the Cell* Training Course, Marine Biological Laboratory — 2017
- Differential Equations of Mathematical Physics, Northwestern University — 2016
- Engineering Analysis 4 (applied differential equations, lead TA), Northwestern University — 2013
- Honors Calculus for Engineers (multivariable calculus), Northwestern University — 2012

Linear Algebra and Differential Equations, University of Illinois at Urbana-Champaign 2007 – 2008

OUTREACH EXPERIENCE

Presented research to underserved high school students through the Schuler Scholar Program 2015 – 2016

Cofounded and led the *Communicating Science-Chicago (ComSciCon-Chicago)* workshop 2014 – 2016

Won 4th place in the Northwestern Scientific Images Contest 2016

Organized public outreach activity *American Scidol* at the MIT Museum 2014

Volunteered at the Northwestern Graduate Leadership Council's Science Pentathlon for middle schoolers 2014 – 2015

Organized the *Communicating Science National Workshop (ComSciCon)*, Cambridge, MA 2014

Attended inaugural *Communicating Science National Workshop (ComSciCon)*, Harvard University 2013

PROFESSIONAL SERVICE

Faculty advisor to Students for the Advancement of Microbiology (SAM) 2024 – present

Organizing Quantitative Microbiology Journal Club for graduate students 2024 – present

Organized *Theory in Biology* virtual seminar 2021 – 2022

Refereed manuscripts for *Nature Microbiology*, *Cell Systems*, *The ISME Journal*, *Nature Plants*, *Science Advances*, *Current Biology*, *iScience*, *SIAM Journal on Applied Mathematics*, *Journal of Mathematical Biology*, *Chaos*, and *Communications in Nonlinear Science and Numerical Simulation* 2013 – present

Designed and administered the *Math and Climate Research Network* online platform 2015 – 2016

Organized the *Mathematics of Climate Tipping Points* focus group 2012 – 2015

SELECTED PRESS

News & Views in *Nature Ecology & Evolution*, Smooth functional landscapes in microcosms October 2, 2023

Dispatch in *Current Biology*, The metabolic rate is the trait March 14, 2022

U. Chicago News, The genomic structure of microbial communities can predict metabolic activity January 26, 2022
Also carried by Northwestern News, Phys.org, and EurekaAlert.

SIAM News, Modeling Vegetation Patterns in Vulnerable Ecosystems March 1, 2017